CHARAN TEJA VALAVALA

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GitHub | Portfolio | LinkedIn

SOFTWARE ENGINEERING | APPLIED AI

SELECTED PROJECTS

Course Copilot | Code | Demo | Live — Python | FastAPI | MiniLM | SQLite

- Built PDF/TXT ingestion, file/page citations and hybrid search with MiniLM, FTS5 BM25 and 2:1 rank fusion; added an automated NLI support gate for optional generated answers.
- Benchmarked fixed retrieval on **300 public SciFact queries over 5,183 abstracts**: Hit@5 **78.7%** vs 76.0% lexical, MRR@5 0.668; separate handbook checks abstained on 8/8 out-of-scope prompts.

FormScope | Code | Demo | Live — Python | FastAPI | ONNX | XBRL

- Built SEC XBRL APIs with exact period/unit matching, restatement handling and source-linked financial ratios; added lexical and quantized MiniLM filing search through ONNX CPU inference.
- Matched **40/40** revenue and net-income values against pinned references from 20 filings; hybrid ranked the labeled excerpt first on 30/30 questions over 20 selected excerpts, a small, biased set.

IncidentPilot | Code | Demo | Live — Python | scikit-learn | Qwen | SQLite

- Implemented read-only investigation tools, evidence consistency checks, versioned SQLite traces and approval-gated simulated actions; compared a TF-IDF/logistic-regression classifier with local Qwen.
- Trained the classifier on 20 synthetic cases; on **8 separate stress cases**, it scored **8/8** vs Qwen candidate scoring **4/8**. Reported diagnosis coverage, fallback and evidence-reference validity.

ACADEMIC PROJECTS

Diabetic Retinopathy Detection — Python | OpenCV | scikit-learn

- Extracted color and texture features from retinal fundus images and trained SVM classifiers; evaluated accuracy, precision, recall and sensitivity.

Reinforcement Learning for Object Tracking — Python | Monte Carlo control

- Built a stochastic grid environment with terminal states and teleportation; compared on-policy and off-policy Monte Carlo control through convergence and policy behavior.

EXPERIENCE & EARLY-STAGE WORK

MDDTI SBC | Web Development Intern — May - July 2024

- Contributed HTML/CSS frontend pages and MongoDB-backed features to the website and online test platform used by **200+ students and trainees**.

DataForge | Founder, concept-stage venture — February - May 2026

- Explored an AI infrastructure and data center venture with mentor guidance and early funding discussions; the venture remained at the concept stage.

EDUCATION & ACHIEVEMENT

BITS Pilani, Hyderabad Campus | B.E. Computer Science Engineering (currently enrolled)

AP EAMCET: Statewide rank 8

Narayana Junior College | Class XII: 98.1% | 2022

Narayana High School | Class X: 10/10 GPA | 2020

TECHNICAL SKILLS & COURSEWORK

Programming: Python, SQL, C, C++, Java, JavaScript, HTML, CSS

Data and ML: NumPy, Pandas, Matplotlib, scikit-learn, OpenCV

AI tooling: Sentence Transformers, Hugging Face Transformers, PyTorch (inference), ONNX, Ollama

Backend and delivery: FastAPI, REST APIs, MongoDB, SQLite, Git, Docker, pytest, GitHub Actions

Coursework: Artificial Intelligence, Machine Learning, Reinforcement Learning, Foundations of Data Science, Data Structures and Algorithms, Database Systems, Object-Oriented Programming

Spoken languages: English, Hindi, Telugu